
Capstone 3 – Fahad Ali

AI-Powered Transaction Intelligence & Financial Coaching System



Business Problem & Context



Consumers generate thousands of financial transactions every year, yet most people struggle to understand their spending behavior.



Current financial tools mainly provide transaction logs and basic summaries but lack predictive insights or proactive guidance.



This creates an opportunity to apply data science to:

- Identify patterns in financial behavior
- Predict spending patterns
- Support future intelligent financial coaching tools

Stakeholders, Success Metrics, and Constraints

Stakeholders

- Individual banking customers
- Fintech platforms and financial apps
- Financial advisors

Success metrics

- Strong predictive accuracy (R^2 score)
- Reasonable prediction error
- Model interpretability

Constraints

- Limited dataset features
 - No external economic variables
 - Time constraints for model optimization
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Framing the Data Science Problem

The project was framed as a supervised regression problem



Goal: predict the transaction amount using demographic, behavioral, and time-based features.



Why prediction matters:

Helps identify unusual spending patterns

Supports financial recommendations

Enables proactive financial tools

Dataset Overview



The dataset contains approximately 50,000 transaction records.



Each record includes:

Customer demographic data

Merchant information

Transaction category

Transaction date and time

Income vs expense classification



Target variable: **transaction_amount**



This dataset represents structured financial transaction behavior similar to what fintech systems collect

Data Wrangling & Preparation



Before modeling, the dataset required several preprocessing steps.



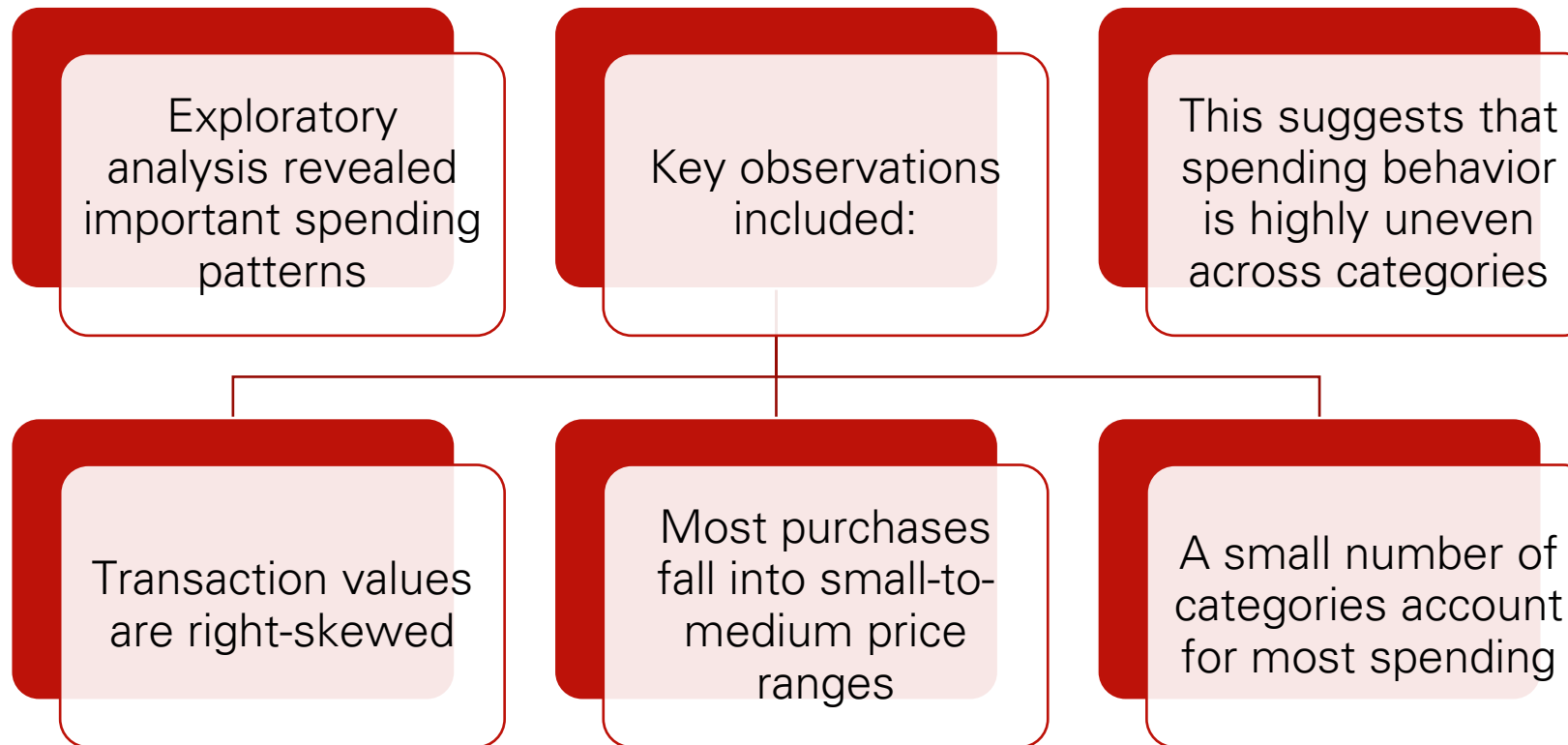
Key transformations included:

- Converting birthdate into age
- Extracting month and day-of-week features
- Encoding categorical variables
- Handling high-cardinality merchant data
- Scaling numerical features



These steps ensured the data was clean, structured, and suitable for machine learning.

Exploratory Data Analysis (EDA)



Spending Behavior Insights



Additional analysis revealed temporal patterns in spending



Key findings:

Spending varies by month and day of week
Certain categories dominate overall spending
Merchant information strongly correlates with purchase size



These insights helped guide the model selection

Modeling Approach



A Random Forest Regressor was selected.



Reasons for choosing Random Forest:

Captures nonlinear relationships

Handles mixed data types well

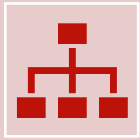
Robust with minimal tuning

Provides feature importance for interpretation



The dataset was split: **80% Training / 20% Testing**

Model Performance



The model produced the following results:

R^2 Score: 0.82

MAE: 24.15

RMSE: 36.90



Interpretation:

The model explains 82% of the variance in transaction amounts

Average prediction error is about \$24 per transaction



Overall, the model captures meaningful patterns in financial behavior.

What Drives Spending?



Feature importance analysis revealed that:

- Transaction category is the strongest predictor
- Merchant information significantly improves predictions
- Temporal features such as month influence spending
- Age has smaller influence



This confirms that spending behavior is primarily contextual rather than demographic.

Business Recommendations



Based on the results, financial platforms could implement:

Behavioral spending alerts: detect when transactions exceed predicted spending patterns

Personalized budget insights: identify categories where users may overspend

AI Financial coaching: translate predictive insights into natural language recommendations



Example: *"Your dining spending tends to increase on weekends. Setting a weekly dining budget may improve savings."*

Practical Considerations & Future Work

This model serves as a baseline system.

Future improvements include:

- Hyperparameter tuning
 - Gradient boosting model comparison
 - Time-series forecasting for spending trends
 - Customer segmentation via clustering
 - Integration with LLM-powered financial coaching
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Conclusion

This project demonstrates an end-to-end machine learning workflow:

- Data preprocessing
- Exploratory analysis
- Feature engineering
- Predictive modeling
- Model evaluation

The result is a predictive analytics system capable of powering intelligent financial tools.

Questions

Thank you!